Forecasting Transportation Cost Trends with Macro factors

A multivariate analysis using **Crude oil price**, **Car loan rates** and **Consumer Confidence**

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Business Problem:
Lack of forward visibility into transportation cost inflation

Detailing the problem

Businesses spend **8-12% of COGS on freight**, still use lagging indicators (last Quarter data) to budget

Budget misses: 5-10% overspending when costs spike

Leads to **Carrier pricing disputes** when no data available to justify surcharges

Inventory misalignment by overstocking when costs are expected to spike

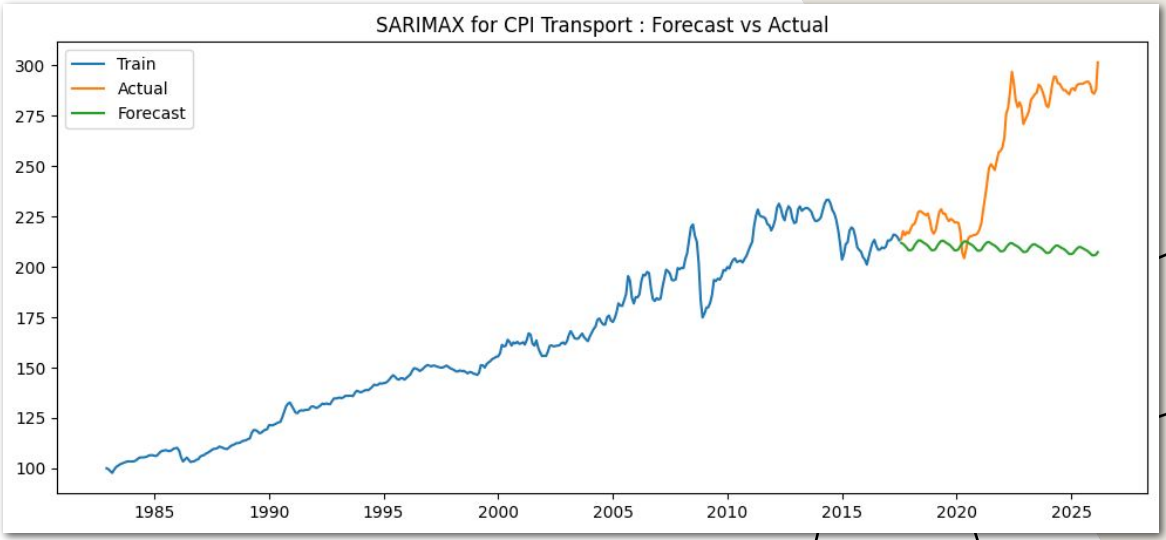
Enter CPIETRANS

Research Consumer Price Index - Transportation published by [BLS.gov]

Contains not only the Fuel Oils CPI but also includes other Trucking costs - labor, maintenance i.e., **Comprehensive**

Forecasting CPIETRANS can help us anticipate transportation costs better

Limitation: Need exogenous variables



Data and Exogenous Variables

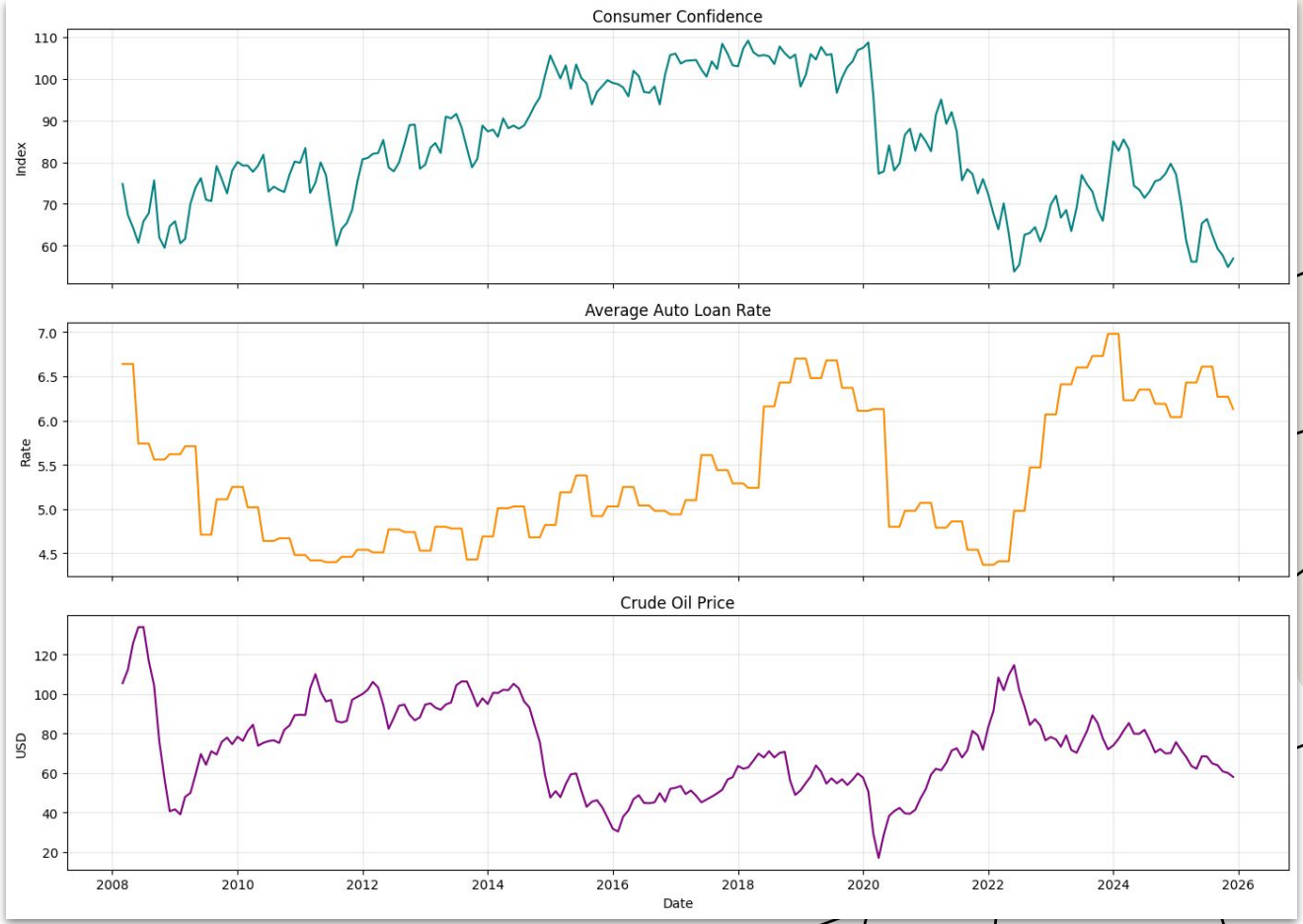
Additional factors outside of the organisation's control but influence

Consumer Confidence drives demand

Average Auto Loan Rate affects new vehicle purchase

Crude Oil Price primary input for fuel cost

All data considered are between Mar 2008 till Dec 2025



Solution

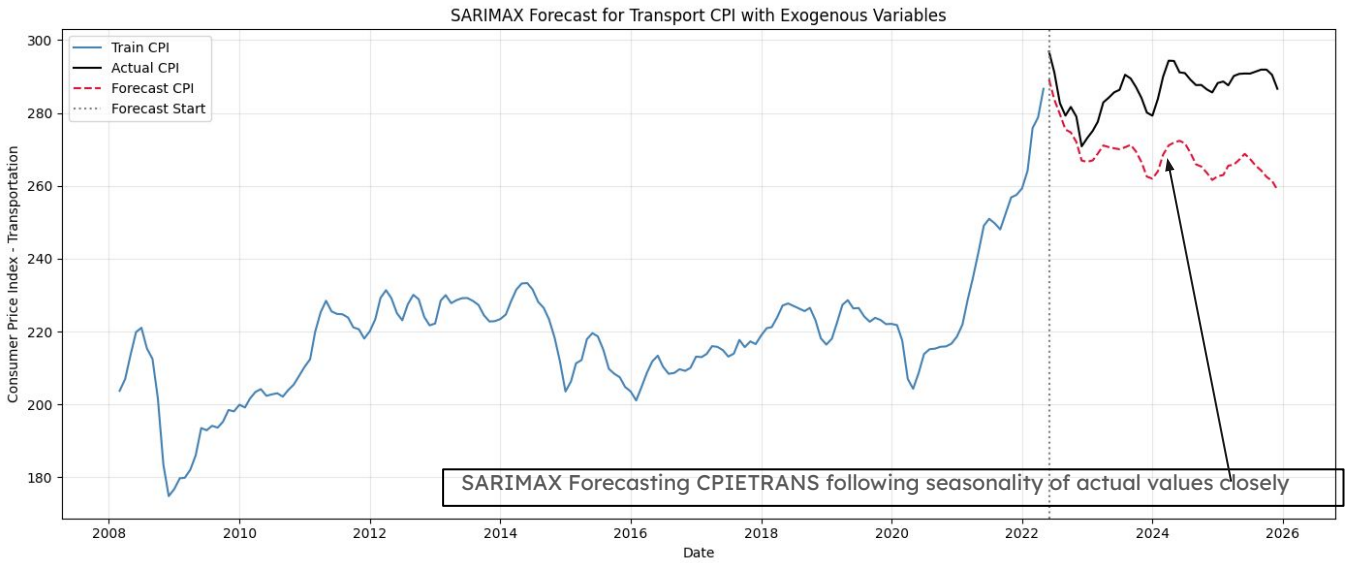
Analyze factors affecting transportation cost :
Target variable - CPI Transportation
exogenous factors - Crude Oil Price, Car Loan rates & Consumer Confidence

Build a multi-variable machine learning model, and forecast Transportation costs

Identifying action items for each sector of the business to mitigate issues

Model Output & Recommendations

Accuracy and error rates



Metric	Value
AIC	-2.12
MAE	3.24
RMSE	4.67
MAPE	1.8%

- AIC score indicates that the **data fits better than the baseline models** (with only Fuel Oil data)
- MAE being 3.24 indicates predictions are **off by 3 CPI points**, fairly small when they are in the range of 250-300
- MAPE is 1.8%, indicates that the forecast is always within 2% of the actual value, thus **accuracy is higher**

Observations from Forecast

1. **Transportation costs are not expected to explode, but they remain elevated.**
   The forecast suggests CPIETRANS stays in a relatively high range rather than returning quickly to the lower pre-spike levels.
2. **The direction is more important than the exact month-to-month wiggles.**
   Business users should focus on the overall path of the forecast, which indicates whether cost pressure is easing, staying flat, or rising.
3. **The model gives a useful planning baseline, not a guarantee.**
   Forecasts are best used as a starting point for budgeting and scenario planning, especially when fuel prices or consumer demand shift quickly.
4. **The forecast is most valuable for decisions that happen before costs are realized.**
   Supply chain teams can use it to time carrier negotiations, set freight budgets, and prepare contingency plans for higher or lower transportation spend.

Key takeaway

transportation inflation is expected to stay significant, so supply chain teams should plan conservatively and use the forecast as an early warning system

Business Impact

- **Budget pressure:** If the forecast points upward, freight and logistics costs may rise, which affects margins and operating plans.
- **Negotiation timing:** If costs are expected to ease, procurement teams can use that signal when negotiating rates or fuel surcharges.
- **Capacity planning:** If transportation inflation is expected to stay high, analysts can prepare for tighter budgets, slower replenishment decisions, or more conservative shipment planning.
- **Risk management:** The model gives an early warning signal so businesses can prepare for scenarios instead of waiting for actual costs to spike

Intended audience - Supply Chain Leaders, Transportation Planners - freight managers, procurement teams, logistics analysts, finance teams responsible for budgeting/negotiating shipping costs

Limitations of model

Depends on future exogenous values

Exogenous relationships can change

Misses sudden shocks

Assumes past patterns remain informative

Sensitive to non-stationary trends

Model choice matters

Accuracy

Not cause and effect

Accuracy degrades over long horizons

May underreact to regime changes

Data quality is critical

Business recommendations - Part 1/2

1. **Set freight budgets using a conservative transport-cost baseline.**
   Because the forecast still points to elevated transportation CPI, budget owners should avoid assuming a quick return to pre-spike cost levels.
2. **Use the forecast to time carrier negotiations.**
   If the model shows easing pressure, procurement teams can push harder on rate discussions and fuel surcharge terms before renewed cost pressure appears.
3. **Build scenario plans around fuel volatility.**
   Since crude oil is an exogenous driver, create best/base/worst cases so planning does not rely on a single forecast path.
4. **Monitor leading indicators monthly.**
   Track oil prices, consumer confidence, and auto/credit conditions every month so the forecast can be refreshed before budgeting cycles.
5. **Use the model as an early warning signal, not a final answer.**
   The forecast is strongest when it helps the business react earlier to cost pressure, not when it is treated as exact truth.

Business recommendations - Part 2/2

What these mean in practice

- **Finance** can use the forecast to set reserve amounts and margin expectations.
- **Procurement** can use it to plan renegotiation windows and surcharge clauses.
- **Operations** can use it to decide when to accelerate shipments or delay non-urgent moves.

Executive takeaway

A simple recommendation line would be: **“Use the CPIETRANS forecast as a planning baseline for transportation spend, then pair it with monthly monitoring and scenario analysis to reduce budget surprises.”**

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